



life.augmented

Releasing your creativity

Discover the STM32 family of
microcontrollers & microprocessors



STM32: a developer-first strategy since 2007



STM32 is a key enabler: empowering embedded developers around the world to release their creativity.

We provide embedded developers with cutting-edge hardware and software technology, comprehensive support, and high-quality, reliable supply. This helps them build designs that are smarter, more connected, and more secure.

**The first choice for
32-bit MCU developers**

Source: Aspentec embedded survey, 2022

#1 GP MCU
Worldwide

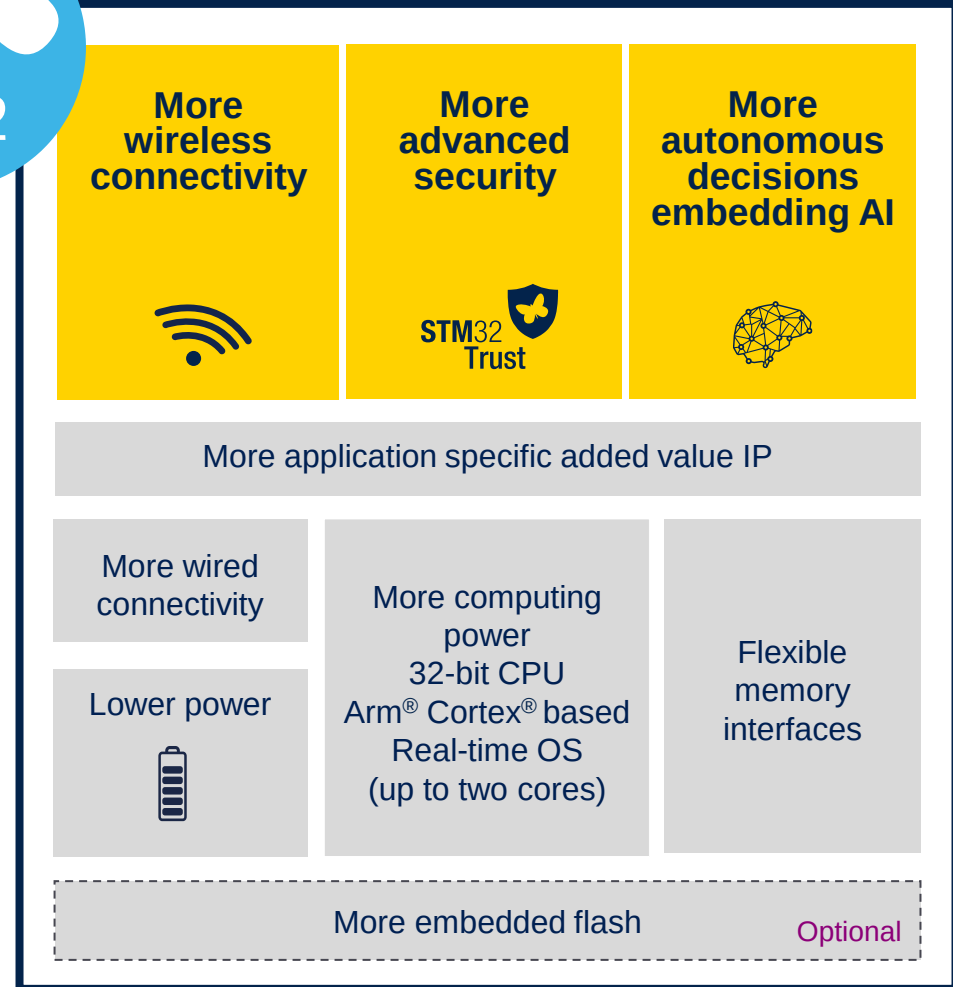
Source: OMDIA CLT, 2022,2023

100,000+ customers

Our technology starts with You



Supporting developers' needs





What the STM32 family offers

Real-time performance

- Powerful Cortex® cores
- Multicore performance
- Fast interfaces
- Hardware accelerators



Outstanding power efficiency

- Ultra-low dynamic power consumption
- Long lifetime, small battery
- Sustainable technology



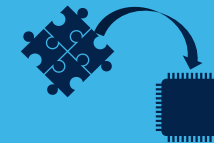
Advanced, innovative peripherals

- Graphic acceleration
- Digital & analog peripherals
- USB Type-C®
- Peripherals for wireless and edge AI solutions



Optimized integration

- Best fit for application requirements (package size, cost, performance)
- Safety & security features



Extensive ecosystem

- Comprehensive development tools
- Wide range of partners
- Community support



3,300+ part numbers



Rolling 10-year longevity commitment for continuous supply



The STM32 portfolio

Five product categories



Wireless
MCU

Short- and long-range connectivity



Ultra-low-power
MCU

32-bit general-purpose microcontrollers: from 75 to 3,224 CoreMark score



Mainstream
MCU



High-performance
MCU



Embedded
MPU

32- and 64-bit microprocessors



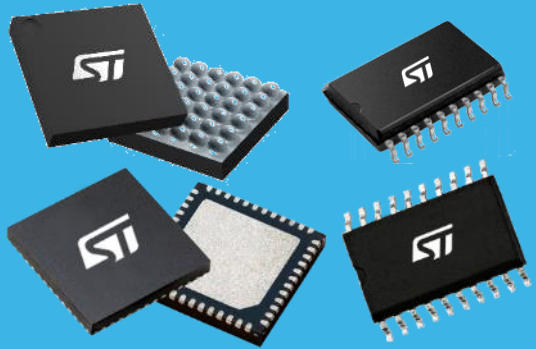
Enabling edge AI solutions



Scalable security



Addressing entry-level to high-performance applications



90+ package types
from 5 to 784 mm²



20- to 68-pin QFN
18- to 208-pin WLCSP
20-pin TSSOP
8-pin SO
32- to 208-pin LQFP
64- to 240-pin(+25) BGA

**3,300+
part numbers**



Multiple memory options

From 8 Kbytes to 4 Mbytes flash memory
From 2 Kbytes to 2 Mbytes RAM

STM32C0

8 pins
16 Kbytes flash
memory
32 MHz

XXS



XXL



STM32H7

240 pins
2 Mbytes flash memory
550 MHz



STM32 portfolio



 MPU						STM32MP1 Up to 1 GHz Cortex-A7 209 MHz Cortex-M4	STM32MP2 Dual 1.5 GHz Cortex-A35 400 MHz Cortex-M33
 High-performance MCUs				STM32F2 Up to 398 CoreMark 120 MHz Cortex-M3	STM32F4 Up to 608 CoreMark 180 MHz Cortex-M4	STM32F7 1082 CoreMark 216 MHz Cortex-M7	STM32H7 Up to 3224 CoreMark Up to 550 MHz Cortex -M7 240 MHz Cortex -M4
 Mainstream MCUs			STM32F3 245 CoreMark 72 MHz Cortex-M4	STM32G4 569 CoreMark 170 MHz Cortex-M4	<i>Mixed-signal MCUs</i>		
	STM32C0 114 CoreMark 48 MHz Cortex M0+	STM32F0 106 CoreMark 48 MHz Cortex-M0	STM32G0 142 CoreMark 64 MHz Cortex-M0+	STM32F1 177 CoreMark 72 MHz Cortex-M3			
 Ultra-low-power MCUs			STM32L0 75 CoreMark 32 MHz Cortex-M0+	STM32L4 273 CoreMark 80 MHz Cortex-M4	STM32L4+ 409 CoreMark 120 MHz Cortex-M4	STM32L5 443 CoreMark 110 MHz Cortex-M33	STM32U5 651 CoreMark 160 MHz Cortex-M33
 Wireless MCUs			STM32WL 162 CoreMark 48 MHz Cortex-M4 48 MHz Cortex-M0+	STM32WB 216 CoreMark 64 MHz Cortex-M4 32 MHz Cortex-M0+	STM32WBA 407 CoreMark 100 MHz Cortex-M33		

STM32 high-performance MCUs





STM32 high-performance MCUs

Up to 3224 CoreMark and a rich set of peripherals

STM32H7

- Dual Arm® Cortex®-M7 + Cortex®-M4 FPU at 480 MHz
- 1327 DMIPS and up to 550 MHz. 1177 DMIPS on single core Arm® Cortex®-M7
- From 512 Kbytes to 2 Mbytes of flash memory
- Very high performance with embedded flash & external memories

STM32F7

- Arm® Cortex®-M7 + FPU at 216 MHz – 462 DMIPS
- From 256 Kbytes to 2 Mbytes of flash memory
- Very high performance with embedded flash & external memories

STM32H5

- Most powerful Arm® Cortex®-M33 MCU yet – 375 DMIPS
- From 128 Kbytes to 2 Mbytes of flash memory
- Industry 4.0 and smart homes

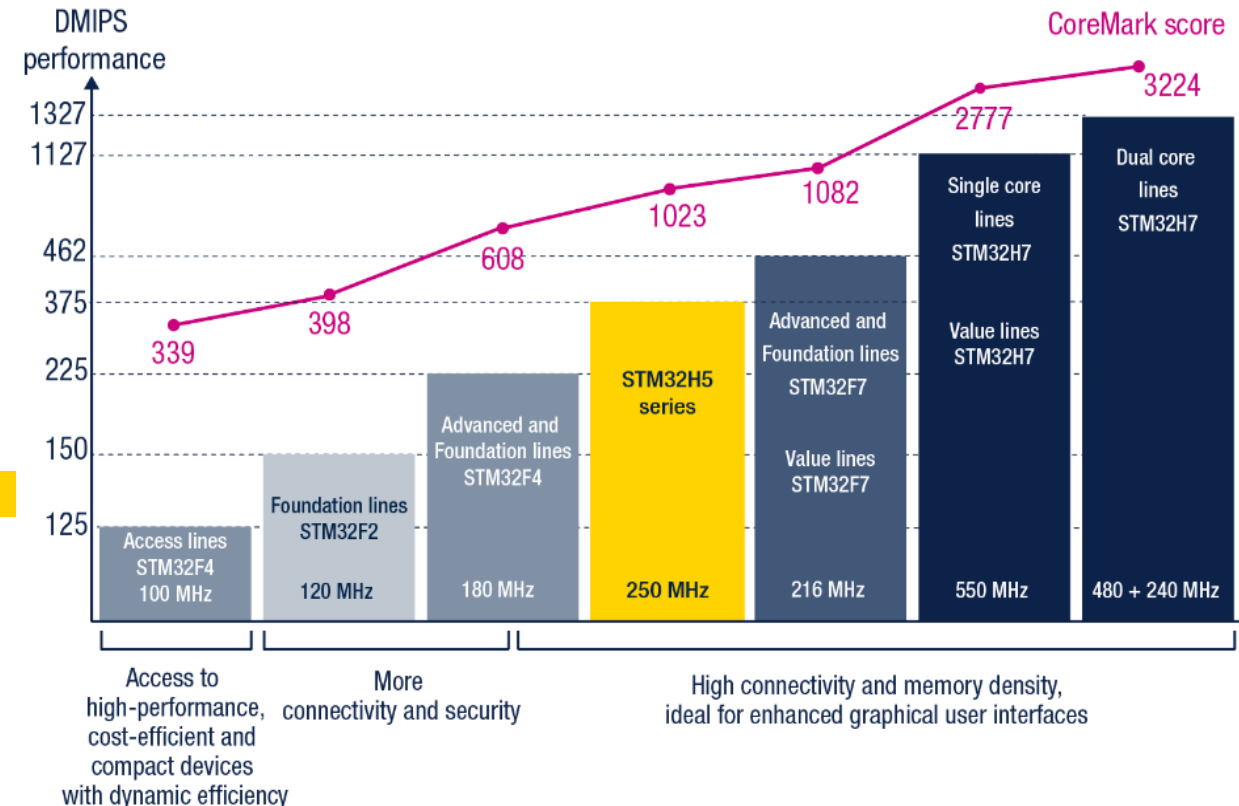
LATEST GENERATION

STM32F4

- Arm® Cortex®-M4 + FPU up to 180 MHz – 225 DMIPS
- From 64 Kbytes to 2 Mbytes of flash memory

STM32F2

- Arm® Cortex®-M3 at 120 MHz – 150 DMIPS
- From 128 Kbytes to 1 Mbyte of flash memory
- Foundation lines for performance and connectivity



STM32N6 preannouncement: more information coming soon.



STM32H5 MCU series for high performance designs



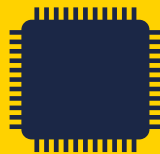
Most powerful Arm® Cortex®-M33 MCU

Industry-first 32-bit MCU with Arm® Cortex®-M33 core running as high as 250 MHz.



Scalable security to address every need

From the most essential security building blocks to fully certified services maintained by ST. First STM32 with TEE.



Optimized cost/performance trade-off

Based on ST's optimized 40 nm process technology.
Large choice of memory, peripherals, and package options.

STM32 mainstream MCUs





STM32 mainstream MCUs

Addressing a large variety of general-purpose applications

STM32F1

- Arm® Cortex®-M3 72 MHz – 61 DMIPS
- Performance and peripherals, user-friendly tools

STM32G0

- Arm® Cortex®-M0+ at 64 MHz – 59 DMIPS
- Maximum IO count per package
- Advanced function: analog, low-power, control

STM32F0

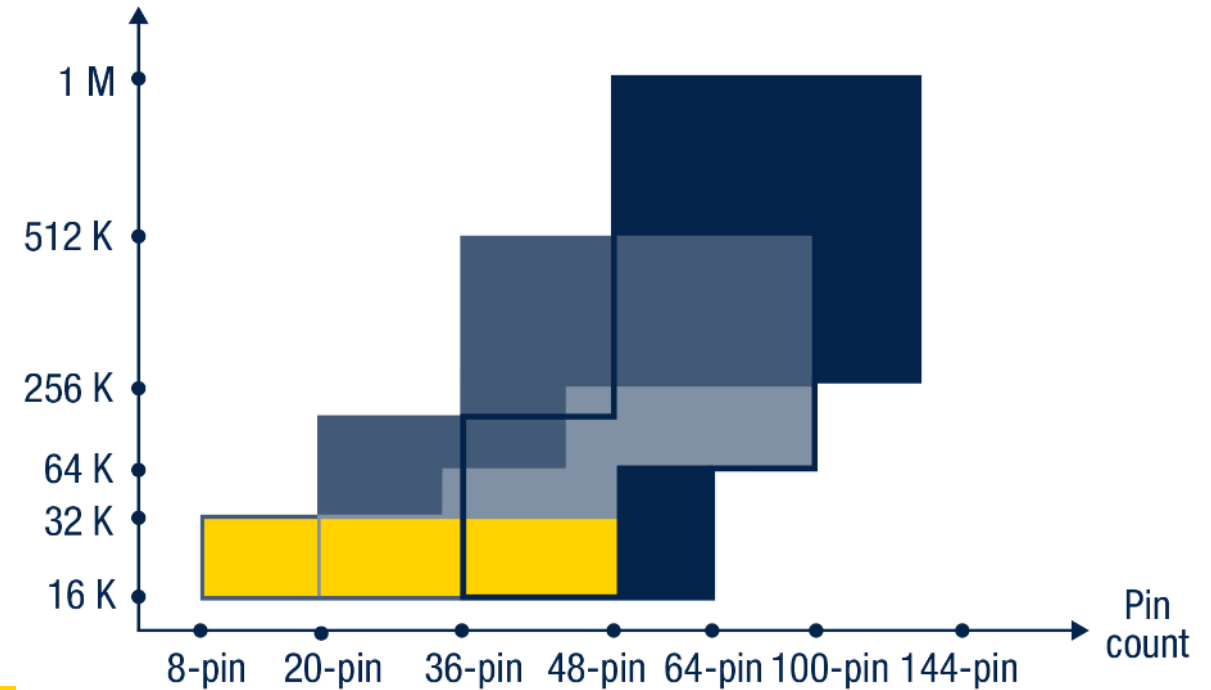
- Entry-level MCU for cost-sensitive operations
- Arm® Cortex®-M0 at 48 MHz – 38 DMIPS

STM32C0

- Arm® Cortex®-M0+ FPU at 48 MHz – 44 DMIPS
- Most affordable 32-bit MCU

LATEST GENERATION

Flash memory size (bytes)





STM32C0 MCU series

Your next 8-bit MCU is a 32-bit!

Streamline costs without compromising performance with ST's most affordable 32-bit MCU.



Affordability

Helps you reduce costs thanks to an attractive price point and an optimized BOM.



Reliability

Benefits from proven STM32 quality & reliability.



Continuity

Consistent pinout with STM32G0 & shares same technological platform.

STM32 ultra-low-power MCUs





STM32 ultra-low-power MCUs

Ultra-low-power, market-proven solutions
150 DMIPS performance

STM32U5

- Arm® Cortex®-M33 + FPU at 160 MHz
- From 128 Kbytes to 4 Mbytes of flash memory
- Lowest power mode + RAM + RTC: 0.35 μ A
- For IoT nodes and graphics



LATEST GENERATION

STM32L5

- Arm® Cortex®-M33 + FPU at 110 MHz
- From 256 to 512 Kbytes of flash memory
- Lowest power mode + RAM + RTC: 0.35 μ A

STM32L4+

- Arm® Cortex®-M4 + FPU at 120 MHz
- From 512 Kbytes to 2 Mbytes of flash memory
- Lowest power mode + RAM + RTC: 0.39 μ A

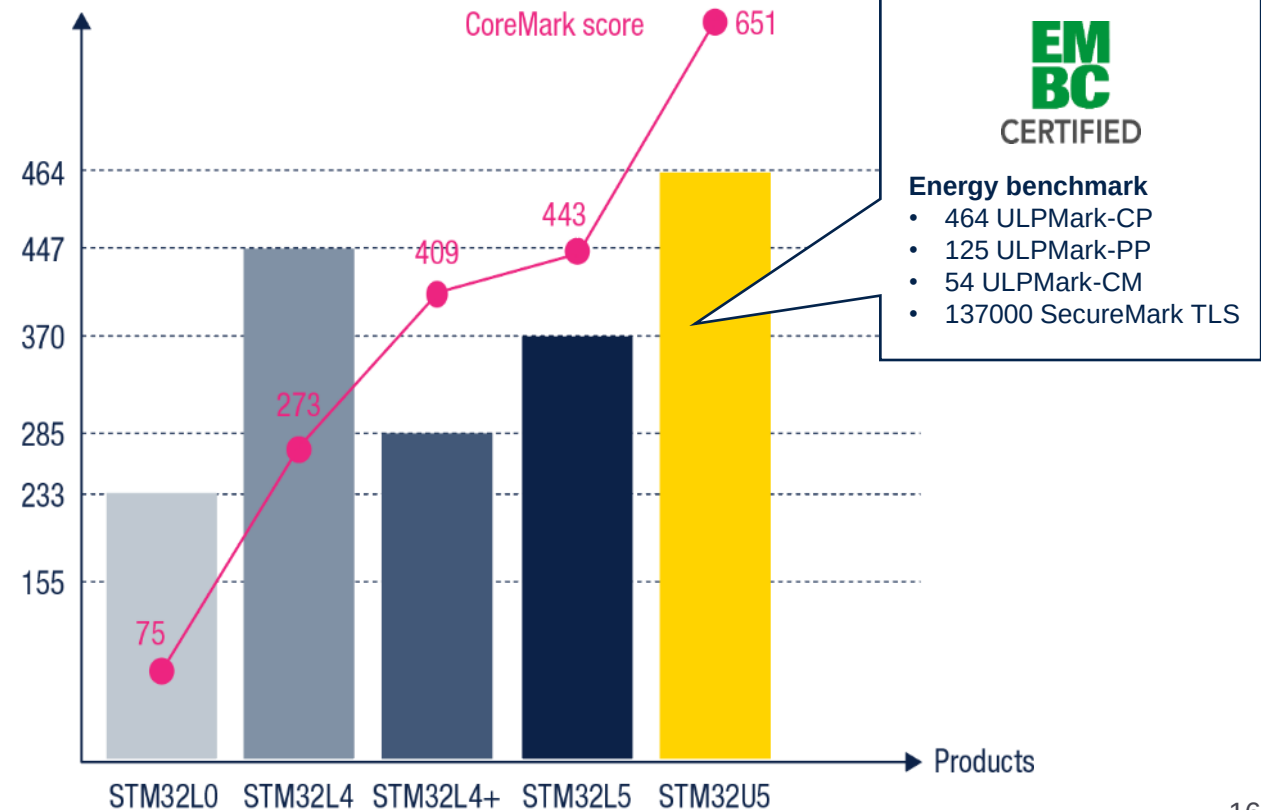
STM32L4

- Arm® Cortex®-M4 + FPU at 80 MHz
- From 64 Kbytes to 1 Mbyte of flash memory
- Lowest power mode + RAM + RTC: 0.34 μ A

STM32L0

- Arm® Cortex®-M0 at 32 MHz
- From 8 to 192 Kbytes of flash memory
- Lowest power mode + RAM + RTC: 0.67 μ A

ULPBench score





STM32U5 MCU series the flagship of ultra-low-power MCUs

For IoT & embedded applications.
Up to 4 Mbytes of flash memory.



**1st MCU
certified by the NIST***



High energy efficiency/integration

Innovative power management features. Low Power Background. Autonomous Mode (LPBAM), DMA, and IP autonomous in LP mode.



High security & safety

AES and PKA, side attack resistant. PSA/SESIP level 3 certified. ECC on flash memory and SRAM.



Enhanced graphic performance

First STM32 with advanced graphics accelerators (ART Accelerator) & NeoChrom GPU based on Arm® Cortex® -M33 running at 160 MHz.

STM32 wireless MCUs





STM32 wireless MCUs

The ideal fit for RF designers looking for more than just a radio device

STM32WBA

- Arm® Cortex®-M33 w/ **TrustZone®** @ 100 MHz
- **1 Mbyte** of flash memory / **128 Kbytes** RAM
- Bluetooth® Low Energy 5.3 (long-range, 2 Mbps, advertising extension)
- Up to **+10 dBm** output power
- Enhanced security



LATEST GENERATION

STM32WB

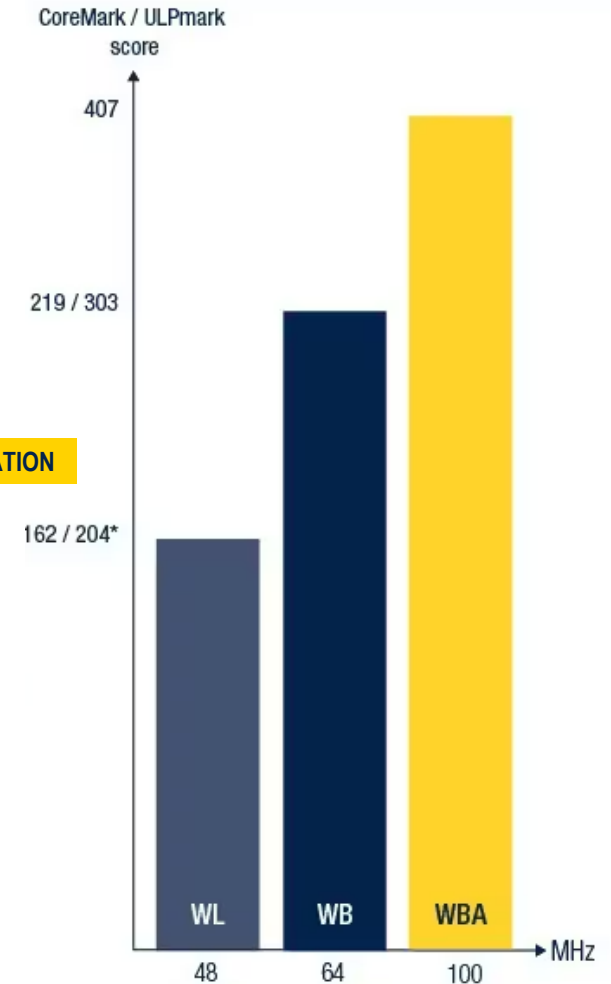
- **Dual core** & security (Arm® Cortex®-M4 /-M0+)
- Up to **1 Mbyte flash- memory/ 256 Kbytes** RAM
- Bluetooth® Low Energy 5.4, Zigbee R22 & Thread, proprietary, Matter Q4'23

OPENTHREAD
released by Google



STM32WL

- World 1st MCU enabling **LoRa®**, **(G)FSK**, **(G)MSK**, **BPSK**
- Arm® Cortex®-M4 and -M0+ at 48 MHz supporting RF – 60 DMIPS
- Up to **256 Kbytes of flash memory / 64 Kbytes** RAM
- Transceiver frequency: 150 to 960 MHz
- Dual-power outputs: up to 22 dBm and up to 15 dBm (Embedded PAs)



Note (*): CoreMark from Flash memory @ 3 V
Pending certification



STM32WBA MCU series: performance, efficiency, and security for the IoT

**Long-range wireless MCU, 2 Mbps, advertising extension
+10 dBm, high security level**



Certified for Bluetooth® Low Energy 5.3

Long range, 2 Msps, advertising extension.
With +10 dBm of output power for better robustness.



Arm® Cortex®-M33 running up to 100 MHz

407 CoreMark score.
100 K cycles for 256 Kbytes of flash memory.



Certified security for faster time to market

Featuring TrustZone® technology.
SESIP level 3 target certification.

STM32 microprocessors



STM32 microprocessors

Making your industrial applications future-proof



STM32MP25

Single or dual Arm® Cortex®-A35 up to 1.5 GHz
Arm® Cortex®-M33 at 400 MHz
NPU at 1.35 TOPS
time-sensitive networking support
3D GPU, 1080p platform

STM32MP2 series

Sampling at OEMs

STM32MP15

Single or dual Arm® Cortex®-A7 up to 800 MHz
Arm® Cortex®-M4 at 209 MHz
3D GPU 720p

STM32MP1 series

Mass market availability

STM32MP13

Arm® Cortex®-A7 up to 1 GHz
Power- and cost-efficient with high security



STM32MP13 MPU lines

Cost-efficient MPUs for industrial and secure applications



Power efficiency

- Best-in-class consumption in low power modes
- Over 90% energy savings in standby and VBAT modes



Certified security services for faster time to market

- SESIP L3 and PSA certified
- PCI ready



Accessible

- Strong, user-friendly ecosystem (OpenSTLinux, Linux-RT, RTOS)
- PCB layout reference designs



STM32MP2 MPU series a step up in performance

Next-gen STM32 MPUs for Industry 4.0 and edge AI solutions

Microprocessors for advanced edge computing

- 64-bit MPU with neural processing unit (NPU)
- 1.35 TOPS (tera operations per second)

Extended connectivity

- Time-sensitive networking (TSN)
- Up to 3 gigabit Ethernet ports (2-port switch)

Advanced multimedia

- 1080p graphics capabilities (3D GPU, H.264 hardware video Codec)
- RGB, MIPI DSI and LVDS displays

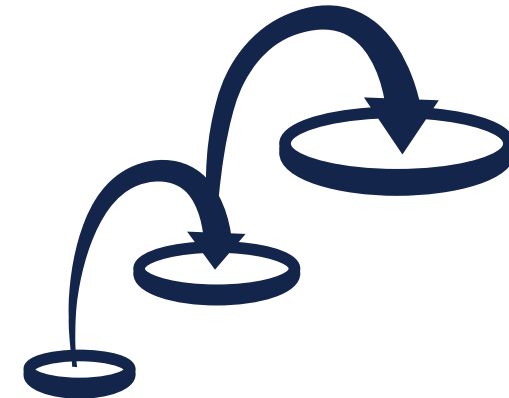
Developer-first strategy: STM32Cube



STM32Cube design ecosystem



**Hardware tools and software helping
you every step of your design journey**





STM32Cube framework

Helping developers release their creativity

Comprehensive offer helping you
accelerate your development

Focus on quality, compatibility, and
stability

Documentations, trainings and
worldwide support channels



Applicative reference implementations

Extension libraries and AI toolkit



Hardware



Embedded
SDK



Development
tool kit



Development
resources





STM32Cube framework

MCU, boards
& software
selection

Hardware and
software
configuration

Application development and
debugging

Code & hardware
programming

Runtime
application
monitoring

STM32
Finder

STM32
CubeMX

STM32
CubeMCU
Packages

STM32
CubeExpansion

STM32
CubeIDE

STM32
CubeProgrammer

STM32
CubeMonitor

Verticals and
partners solutions

IDEs from Partners

Programmers from Partners



Worldwide support channels

STM32 hardware evaluation tools

Easy prototyping, accurate evaluation, and board design references

\$10 → \$30*



STM32 Nucleo boards

Flexible prototyping

70+ references

\$10 → \$100*



Discovery kits

Evaluating key features

40+ references

\$100 → \$500*



Evaluation boards

Full feature evaluation

25+ references



Expansion boards

Add-on functionalities

100+ references



Accessories



Partner boards

From full evaluation to open hardware

20+ references



STM32CubeMCU Packages

Efficient and flexible access to the MCU features

LL drivers	HAL drivers
Lower abstraction level	Higher abstraction level
Lower code size	Higher portability and reuse
MISRA C compliant, statically analyzed, rigorously tested	
Large set of production-ready examples	
Available from st.com , GitHub, or STM32Cube tools	



STM32CubeMCU Packages

Faster development with an optimized and ported selection of market-reference middleware stacks

Middleware	Expansions
AzureRTOS ThreadX and FreeRTOS™ AzureRTOS USBX With support of audio, CDC, HID, DFU, PIMA, Printer, and storage host and device classes AzureRTOS NetXDuo With support of TCP, UDP, IPv4, IPv6, http, MQTT, LWM2M, FTP, PPP, SMTP, and telnet FileX and levelX USB PD and open bootloader Secure boot, Secure Manager API	TouchGFX graphics solution, Motor control, Artificial intelligence MEMS and sensors Secure cloud connectors Functional Safety self-test library
A large set of applicative examples	
Available from st.com, GitHub, or STM32Cube tools	

STM32 Developer Zone for MCUs & MPUs

**Everything for STM32 MCU developers,
in one place**

**A user-friendly environment to help developers
every step of their design journey**

**Direct access to products, hardware and software
tools, embedded software, developer resources**



[STM32 MCU Developer Zone](#)

[STM32 MPU Developer Zone](#)



[Watch the short video](#)

A growing base of partners addressing customer challenges



 Software development tools	 Training
 Hardware development tools	 Engineering services
 Embedded software	 Design houses
 Evaluation boards	 Global services
 Development boards	
 Companion devices	
 Hardware integrated devices	



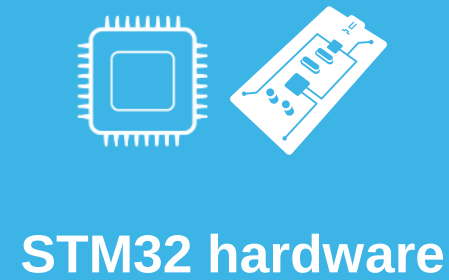
Click to discover
our partners

Solutions with STM32





Helping you build advanced HMIs with a comprehensive STM32 graphics offering



TouchGFX

STM32 software

GUI development tools, GUI code
Examples, library of graphical assets



Introducing NeoChrom GPU

NeoChrom GPU offloads the CPU from the graphic computations, freeing up the memory and boosting performance. Fully supported in [X-CUBE-TOUCHGFX](#).

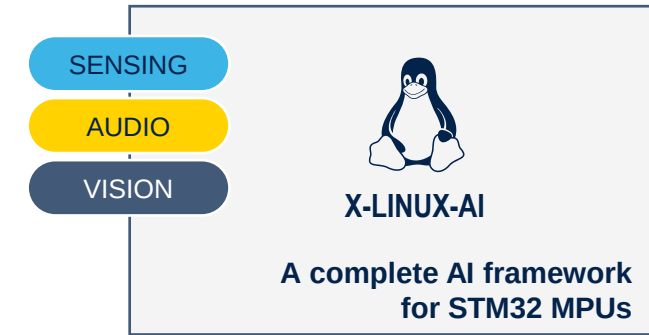
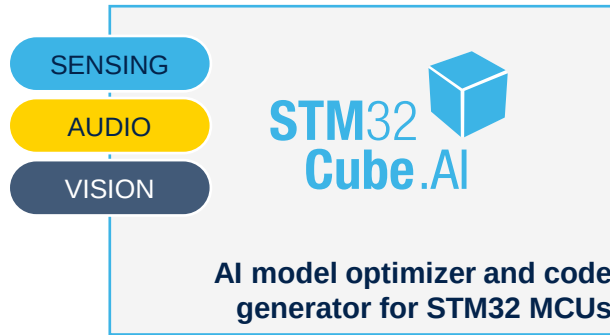


[Watch demos, tutorials and more](#)

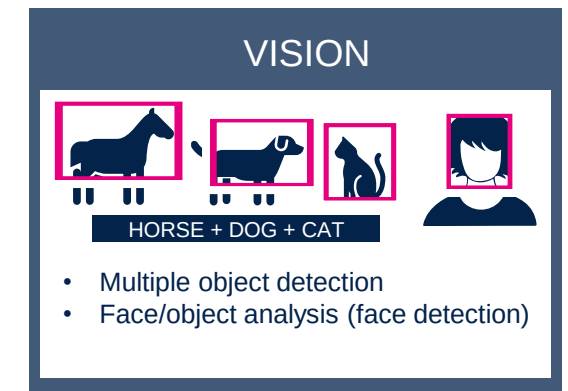
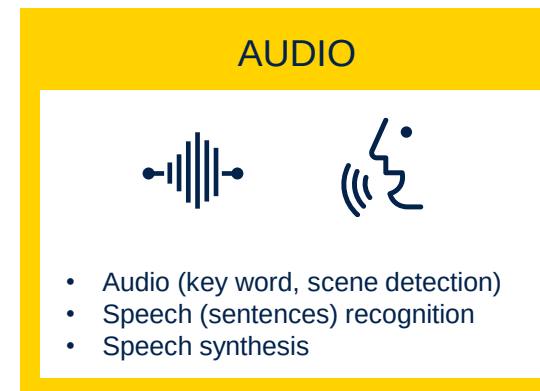
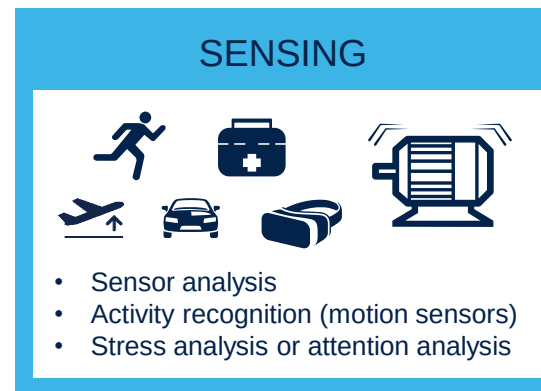
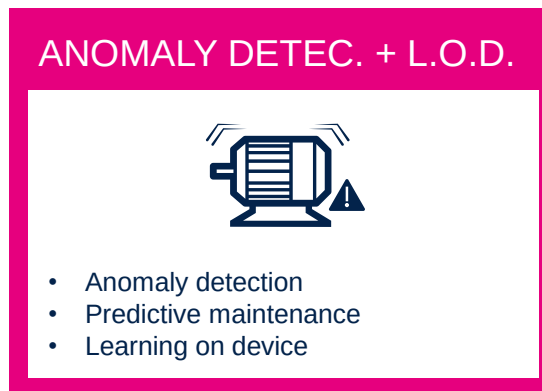


Making edge AI more accessible with STM32 solutions

- 3 products for embedded developers and data scientists



- Covering a broad variety of applications





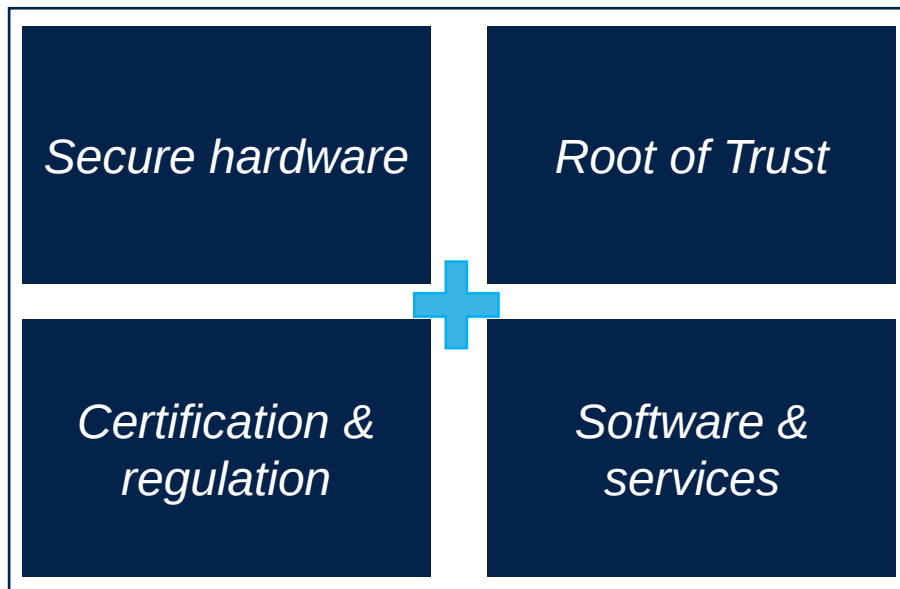
Fast-track your certification journey to meet functional safety standards with STM32

ST provides certified **functional safety packages** and documentation based on robust built-in MCU/MPU safety features.

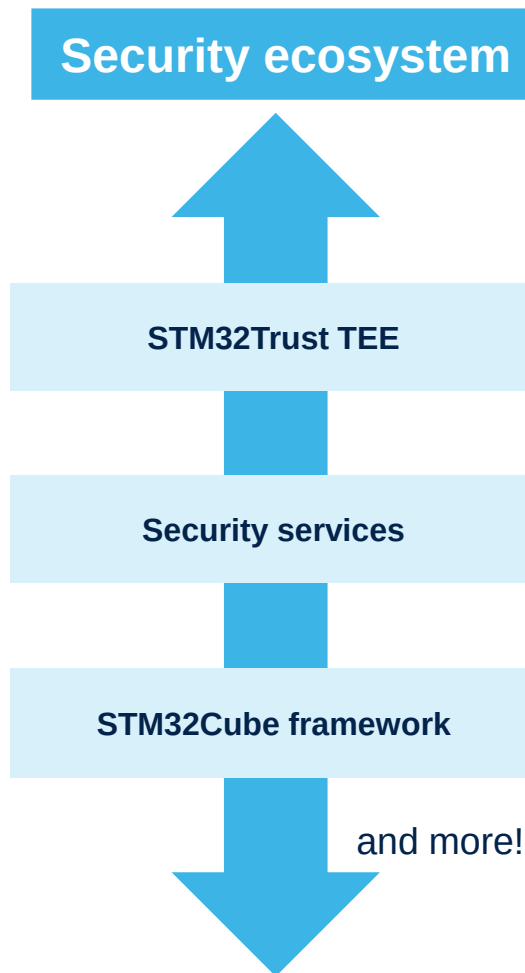
- **SIL functional safety package**
for industrial IEC 61508 (STM32)
- **ASIL functional safety package**
for automotive ISO 26262 (STM8A)
- **Class B functional safety package**
for household electrical appliances
IEC 60335-1/60730-1 (STM32 & STM8)



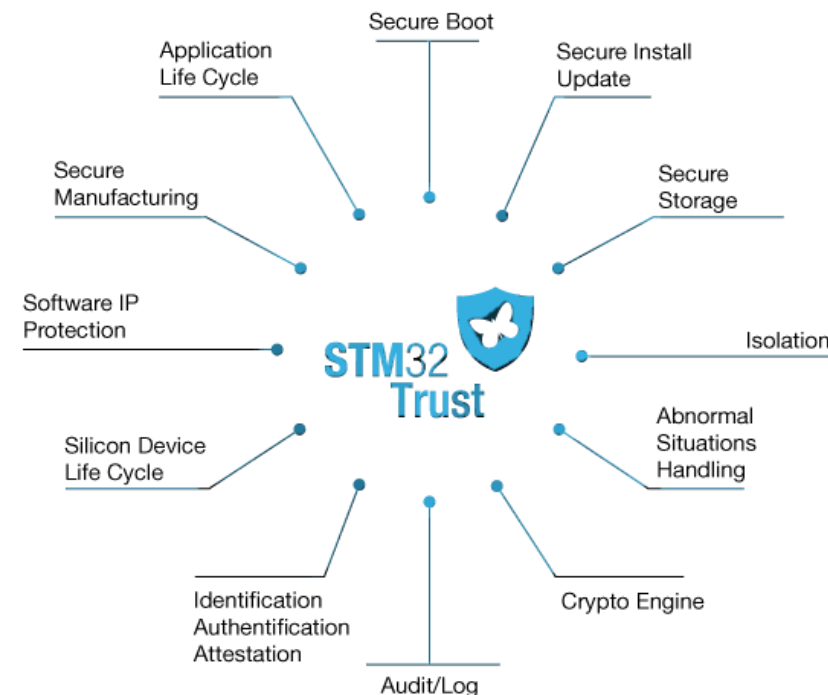
Building trust in embedded systems: the pillars of STM32Trust



Security ecosystem



Security in STM32



Provide the right levels of security assurance thanks to the STM32Trust security functions



STM32Trust – target certifications

 MPU		PSAL1 STM32MP15	PSAL1 SESIPL3 STM32MP13
 High-performance MCUs		PSAL1 STM32H7	PSAL3 SESIPL3 STM32H5
 Mainstream MCUs	PSAL1 STM32G0	PSAL1 STM32G4	PSAL1 STM32C0
 Ultra-low-power MCUs	PSAL1 STM32L4/L4+	PSAL1 SESIPL3 STM32L5	PSAL3 SESIPL3 STM32U5
 Wireless MCUs			PSAL3 SESIPL3 STM32WBA52



Motor control with STM32

Ease STM32 adoption for motor control

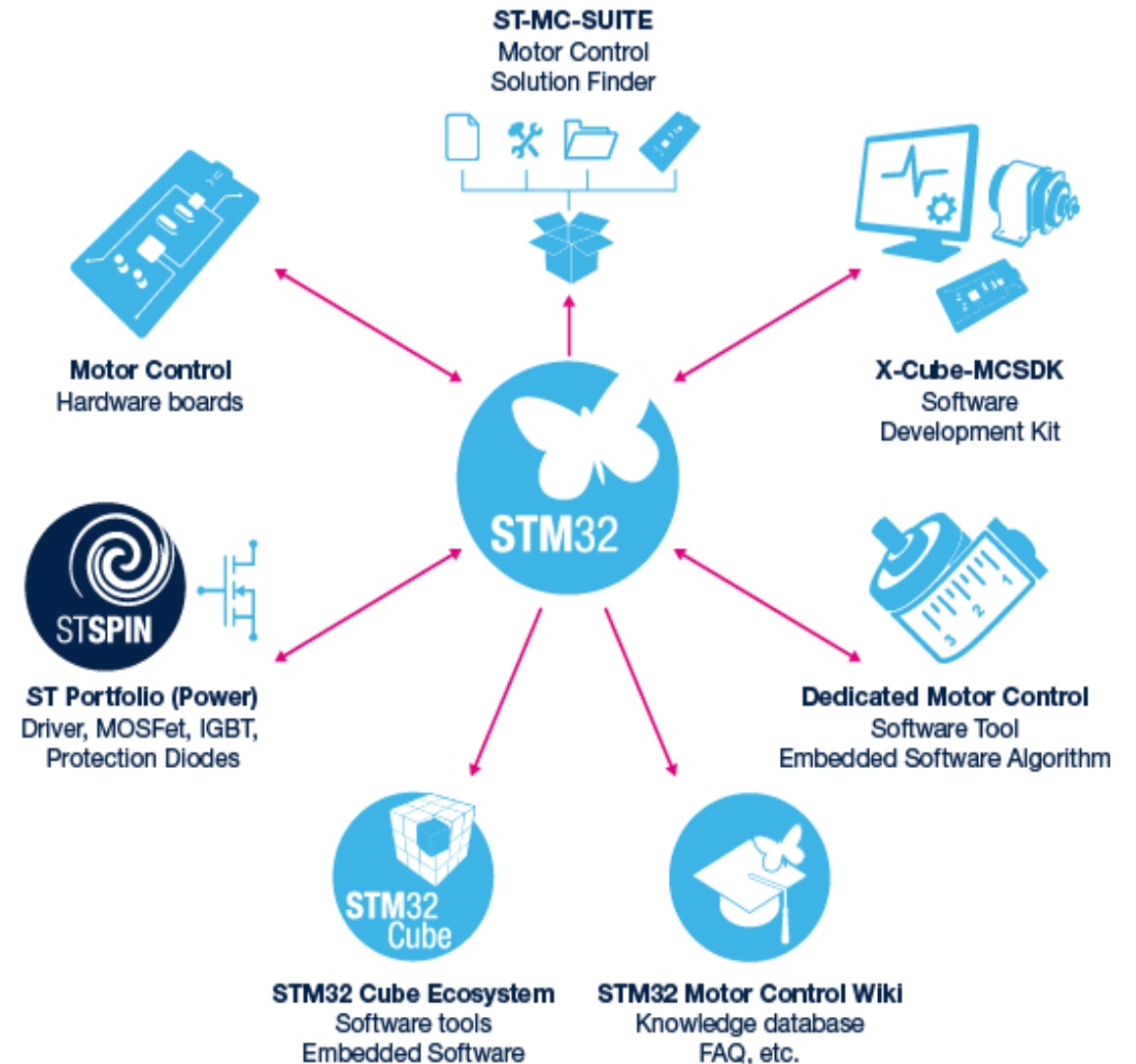
Providing development platform: MC-SDK (firmware library + workbench), MC pilot, MC profiler, hardware boards, documentation.

Innovative products/peripherals and software algorithms

- Advanced motor control timer
- Rich and advanced analog peripherals embedded in the STM32
- Motor profiler

Leverage ST portfolio

Large choice of power components and STM32 to create end-to-end motor control solutions.



Digital power with STM32

Ease STM32 adoption for digital power converters

Development platforms:
DP SDK (PFC and PSU topology examples generator, firmware lib), hardware boards, docs, development tools.

Innovative products/peripherals and software algorithms

- High-resolution timer supporting numerous digital power topologies
- Rich and advanced analog peripherals embedded in STM32
- Hardware coprocessor usage
- Biricha method implementation (ST Authorized Partner)

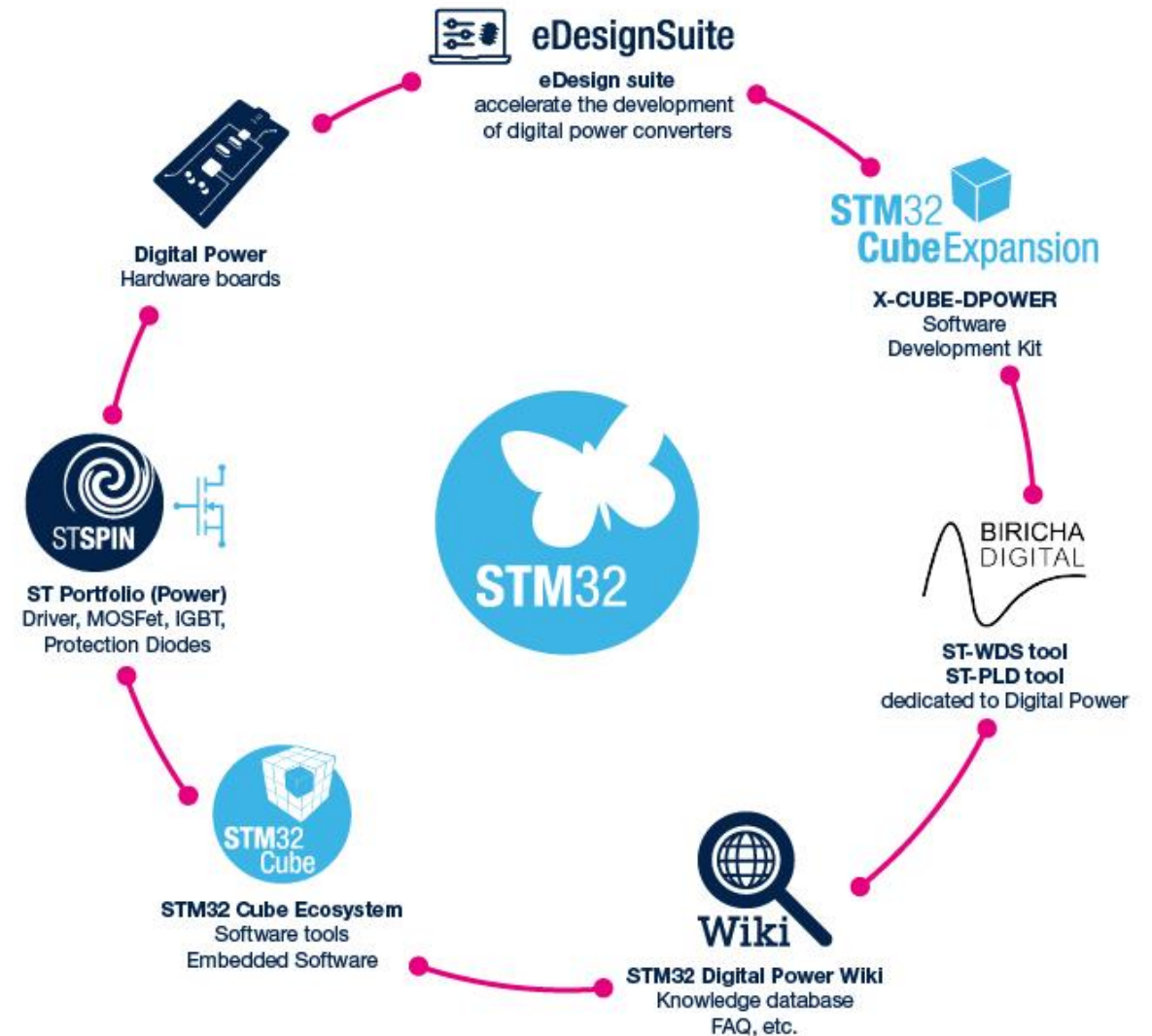
Leverage ST portfolio

Large choice of power components and STM32 to create end-to-end digital power solutions.

PFC and PSU within STM32CubeMX

Firmware pack importation with PFC and PSU topologies implementation in voltage or in current mode running on ST boards.

LATEST NEWS



[STM32 for digital power](#)



USB-C® and power delivery with STM32

More than 560 STM32 MCUs feature a certified USB Type-C® and PD3.1 controller



STM32 supports the latest USB-C® and PD3.1 standards

- SPR and EPR power range up to 240 W, PPS ready
- Sink, source, dual-role power and data roles



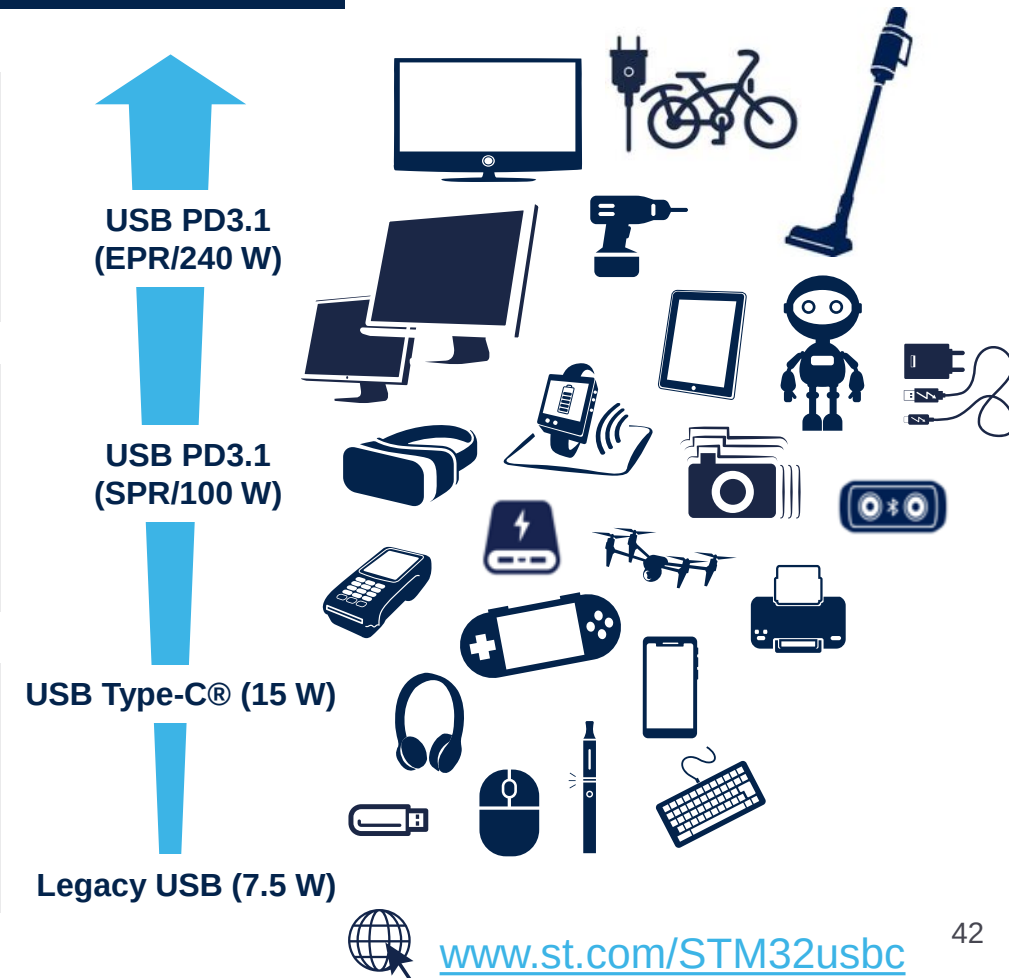
Optimize bill of material and safety

- CC logic, PD transceiver PHY, USB2 device/host interface
- Companion type-c port protection devices (TCPP0x)



Flexible solution and fast prototyping

- Ready-to-use hardware and firmware examples
- Easy debug with STM32CubeMonUCPD software tool



Resources to move forward with your design

Thousands of developers are using STM32!
Join them, share insights, and accelerate your design.



FIND INSIGHTS



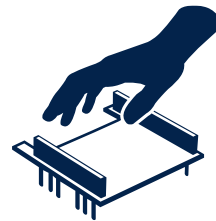
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[STM32 MCU Wiki](#)



[GitHub–STMicroelectronics](#)
[GitHub – STM32 hotspot](#)



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Our technology starts with You



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